



Fig. 3: (a) Jackery 500, a 500W Power Station, (b) EcoFlow Delta 1300, a 1300W Power House, (c) Anker 200, a 200W Power Station, and (d) Bluetti EB70S 800W, 716Wh Power Station

5. DC-DC converters connected to various DC output ports (USB-C, USB, 12V car socket, wireless charging ports)

6. DC-AC inverters producing stable and pure sine wave AC output at its various ports

In Fig. 4, all the major elements of a typical PPS have been shown. The input and output ports are shown in light-green boxes, input electronics in light-blue, output electronics in amber, and the battery and its related electronics has been shown in light-purple background. The design of the unit has following important features:

1. Input-output ports
2. Battery and BMS
3. Enclosure
4. Indication and control
5. Power conversion

Input output ports

The first and foremost design aspects are the type of input and output ports the product should have. Any decent power station should have enough input and output ports

to be able to support multiple charging and discharging devices.

Input ports. A typical input port is the 3-pin AC wall socket. Since portability of the unit is important, it would be good to have a removable AC power cord, which connects to a 3-pin AC input port on the PPS. Better still would be a retractable AC input cord that folds into the unit for easily carrying the same with the unit.

Since the power station may be used away from home/office, it would be great if it could be charged using a solar PV input. The typical 12V input socket from a solar PV panel is a good candidate in this case. Sometimes, the PPS may need to charge from another PPS; in this case there should be a way to charge using direct DC connection between the two PPS units.

Output ports. Output ports may also be the regular 3-pin and 2-pin AC sockets. These would require a good quality inverter—if possible, a 3-phase 220V AC inverter. This is important because some items

may require 120V AC (which can be provided by using a phase and the neutral), and some may require 220V AC (which can be provided across two phases).

While such AC sources are used in homes and offices, one has to remember that unless the PPS unit is connected to the main grid neutral at all times, the AC outputs produced by the PPS will not be ground referred. Therefore, typical earth leakage/fault protection mechanisms (such as residual current breakers) made mandatory in home wirings will not be possible for these AC sources. The PPS may allow the user to connect the true earth ground to the PPS using a special ground terminal, and in this case the PPS must provide the above protections on all its AC outputs.

Typical DC output ports will constitute USB-A ports, USB-C PD ports, 12V jacks, and 12V car charger sockets. A USB-A port may be created by using a simple 5V buck converter of required current capacity. A special power delivery (PD) USB-C adaptor will be required for the USB-C ports. Car charger sockets can provide up to 4A current, and so the 12V buck adaptor used by the PPS must provide that current.

Some PPS units provide an ambient light source with brightness level controls. These are great for outdoor use and use as an emergency light during power cuts.

As wireless charging of smart-phones comes of age and becomes more common, many of the PPS makers would need to provide at least one, and sometimes two, wireless chargers integrated into the PPS. This is a big feature for the consumers who opt for wireless charging of their phones, and would definitely be a great addition to the PPS feature lists. Although, this feature would also increase the cost of the unit quite a bit.